

SunMint — Live Data & Systems Briefing (for partner sharing)

Prepared 2026-08-31 · TrueSight DAO · Sophia Truesight

A briefing for prospective partners: the live systems, the real data caches, how carbon metrics are produced — **and what we are asking from you.**

1. Live public surfaces (production, all open source)

Surface	URL	What it shows
Impact map (PROD)	https://truesight.me/sunmint.html	Registered trees, plot polygons, satellite basemap + dated Sentinel-2 history strip
SunMint whitepaper	https://truesight.me/sunmint/whitepaper/	Methodology (VM0047 / PM002), data model, on-chain carbon lifecycle, carbon formula
Tree monitoring app	https://sunmint.truesight.me/monitor-tree-growth/	Farmer-side measurement flow (photo + calibration card → DBH)

Open-source note: all repositories are **public** on github.com/TrueSightDAO

(`sunmint`, `sunmint_beta/prod`, `truesight_me`, `dao_protocol`, `sunmint_mobile`, etc.).

Every system below can be **independently verified and audited** from source — no black boxes.

2. Raw data caches (machine-readable, live)

Cache	URL	Contents
Tree registry	https://raw.githubusercontent.com/TrueSightDAO/sunmint/main/trees/index.geojson	Every registered tree: id, species, status, coordinates, plot link
Plot registry	https://raw.githubusercontent.com/TrueSightDAO/sunmint/main/plots/index.geojson	Plot boundaries (GeoJSON polygons): name, hectares, status, boundary authority
Satellite cache manifest	https://raw.githubusercontent.com/TrueSightDAO/sunmint/main/satellite/manifest.json	Cached Sentinel-2 scenes per cell/plot with dates + cloud cover
Satellite scenes	https://github.com/TrueSightDAO/sunmint/tree/main/satellite	The actual preview images (<code><lat>_<lng>/<scene-date>.jpg</code>)
Growth measurements	Tree Growth Measurements tab, SunMint spreadsheet	DBH, biomass, CO2e, GPS, signature, timestamps

Data-publication status (honest): the tree/plot/satellite registries above are **already public JSON on GitHub**. The ****growth-measurement** sheet is not yet publicly link-shared** — this is being fixed as part of the verifiable-data work below. Until then, measurement data is reproducible from the open-source code + signed rows, and available on request.

3. Verifiable data — in progress (what we are building next)

We are implementing an **open, verifiable audit trail**:

- **Emit all SunMint RSA signatures as a public JSON payload on GitHub** (every planting event, growth measurement, and plot boundary signed by the submitter's RSA key → a machine-readable `signatures.json` in the sunmint repo, keyed by event/message ID).
- Anyone can then **verify any measurement independently**: signature → public key → payload → the same formula → the same CO2e number.
- This complements the **on-chain audit trail** (planted → measured → credit issued → sold → farmer paid) with a fully open, offline-checkable copy.
- **Public link-share of the measurements sheet** will be enabled as part of this work.

4. How carbon metrics are produced (the formula)

Every tree's CO2e comes from a measured trunk diameter (DBH) via the standard tropical allometric chain:

- **AGB = $a \times \text{DBH}^b$** — above-ground biomass (kg), species allometric (Chave et al. 2014)
- **BGB = $\text{AGB} \times 0.32$** — root biomass (IPCC root:shoot)
- **C = $(\text{AGB} + \text{BGB}) \times 0.47$** — carbon (IPCC fraction)
- **CO2e = $\text{C} \times 44/12$** — CO2 equivalent (molar ratio)

Plus project-level 10% achievement reserve + 20% risk buffer.

One-liner: "We measure each tree's trunk diameter from a photo with a calibration card, convert it to biomass using the standard tropical allometric equation (Chave et al. 2014), then apply IPCC factors — so every tree has a verifiable, method-backed CO2e number."

Methodology alignment: **VM0047** (area-based, VCS module) and **PM002/ACORN** (existing-tree reform) — see the whitepaper §Methodology.

5. Honest note on pre-visit data

No exact site data exists before a visit — and that is **by design**:

- Trees enter the registry **only when planted and signed on-site** (GPS + RSA signature + photo evidence). We do not fabricate standing stock.
- Existing trees on a prospective site are **baseline, not additional** — only reform gains are creditable (the methodology crux).
- The satellite layer monitors **registered plots**; a new site starts with a boundary walk (or CAR/INCRA polygon), then plot-level Sentinel-2 caching begins automatically.

6. Open source — what a partnership really unlocks

Because the technology and data are **open source**, a partnership is not about access to proprietary systems. It is about **joint execution** on:

- **Method-backed carbon metrics (VM0047/PM002)** per tree and per plot
- **On-chain audit trail**: planted → measured → credit issued → sold → farmer paid
- **Satellite change-over-time evidence** for VVB verification
- **Live plot + tree dashboards** on the public impact map
- **Fully open verifiable signatures** (see §3) — anyone can check any number

7. What we are asking from you (the partner)

1. **Guidance on filing the PDD** with the right registry — which VCS / Plan Vivo / registry pathway fits this project best, and how to position the first small-scale pilot for certification success.
2. **Inputs on what the farmers still need to do** — a ground-truth review of our measurement protocol, data completeness, and any missing farmer-side steps before the first verification.
3. **Outreach to potential credit buyers** — introductions/leads for offtake of the VCU once issued, and any buyer-side requirements we should meet now.

8. A shared constraint: small-scale farmers

An important context for how we structure this partnership: our farmers are

****small-scale producers with household income of roughly USD 20,000–30,000 per year****. Any collaboration must:

- Keep farmer onboarding **simple, low-cost, and low-tech** (the app is free,

offline-first; calibration cards are printable locally)

- Ensure **measurable income uplift** for the farmers early (measurement → payment loop), not just at certification
- Respect that their time is limited — field protocols must be quick
- Design cost recovery so certification/registry costs do not fall on the farmers themselves

We invite you to state your needs to be part of this partnership — what would you require (data access, reporting cadence, exclusivity, governance role, cost-sharing) to join? We will answer each point.

Questions welcome. All URLs above are production, open source, and current as of 2026-08-31. Growth-measurement public sharing and the signatures JSON are in progress (see §3).